$$\boldsymbol{H}_v = [\boldsymbol{x}_v, \tilde{\boldsymbol{x}}_v, \boldsymbol{x}_{vr}] \in \mathbb{R}^{3\times L_f}, \quad \boldsymbol{x}_{vr} = \boldsymbol{x}_v - \tilde{\boldsymbol{x}}_v$$

$$\boldsymbol{Q}^{(s)} = \boldsymbol{W}_Q \bar{\boldsymbol{x}}_v^{(s)}, \quad \boldsymbol{K}^{(s)} = \boldsymbol{W}_K \tau(\boldsymbol{c}), \quad \boldsymbol{V}^{(s)} = \boldsymbol{W}_V \tau(\boldsymbol{c})$$

$$\boldsymbol{A}^{(s)} = \mathrm{Softmax}\left(\frac{\boldsymbol{Q}^{(s)}(\boldsymbol{K}^{(s)})^\top}{\sqrt{d_k}}\right)\boldsymbol{V}^{(s)}, \quad \mathrm{MHA}(\bar{\boldsymbol{x}}_v^{(s)}, \boldsymbol{c}) = \boldsymbol{W}_o[\boldsymbol{A}_1^{(s)}; \ldots; \boldsymbol{A}_{N_h}^{(s)}]$$

$$L^{(s)} = \mathbb{E}_{s\sim[1,S], \boldsymbol{x}_v^{(s)}, \epsilon^{(s)}}\left[\mathrm{Huber}(r^{(s)})\right], \quad r^{(s)} = \left\|\epsilon^{(s)} - \epsilon_{\theta_d}(\bar{\boldsymbol{x}}_v^{(s)}, s, \boldsymbol{c})\right\|_2, \quad \mathrm{Huber}(r^{(s)}) = \begin{cases} \frac{1}{2}(r^{(s)})^2, & r^{(s)} < \lambda \\ \lambda(r^{(s)} - \frac{\lambda}{2}), & \text{otherwise} \end{cases}$$